

Test Case

A **test case** is a documented set of conditions, actions, inputs, and expected results used to verify that a software application works as intended. Test cases help testers confirm that the system meets its requirements and identify defects before the software is released.

A well-written test case provides enough detail so that another tester can follow the same steps and reproduce the same result.

Components of a Test Case

The main components of a test case include:

1. Test Case ID

A unique identifier assigned to each test case. This helps testers track, organize, and reference the test case easily.

Example: TC_LOGIN_001

2. Test Description

A brief explanation of what the test case is checking.

Example: Verify that a user can log in with a valid username and password.

3. Preconditions

The conditions that must be met before the test can be executed.

Example: The user must already have a registered account.

4. Test Steps

A clear, step-by-step list of actions the tester must perform.

Example:

1. Open the login page.
2. Enter a valid username.
3. Enter a valid password.
4. Click the Login button.

5. Test Data

The input values required to complete the test.

Example:

Username: testuser@example.com

Password: Test@123

6. Expected Result

The result that should occur if the application works correctly.

Example: The user should be redirected to the dashboard.

7. Actual Result

The actual outcome observed during testing. This is completed after the test is executed.

Example: The user was redirected to the dashboard.

8. Status

The final result of the test case, usually marked as **Pass** or **Fail**.

Example: Pass

Well-Constructed Test Case

A well-constructed test case should be clear, complete, and easy to follow. It should include a specific objective, detailed steps, realistic test data, and measurable expected results.

A good test case should:

1. Use clear and specific language.
2. Include all necessary preconditions.
3. Provide steps in the correct order.
4. Make each step simple and focused.
5. Include realistic test data.
6. State the expected result clearly.
7. Be easy for another tester to repeat.

Well-written test cases improve the quality of testing because they reduce confusion and make it easier to identify defects.

Example of a Well-Constructed Test Case

Field	Details
Test Case ID	TC_LOGIN_001
Test Description	Verify that a registered user can log in using valid credentials.
Preconditions	The user has an active account. The login page is available.
Test Data	Email: testuser@example.com Password: Test@123
Test Steps	1. Open the application login page. 2. Enter the valid email address. 3. Enter the valid password. 4. Click the Login button.
Expected Result	The user is successfully logged in and redirected to the dashboard page.
Actual Result	The user is redirected to the dashboard page.
Status	Pass

Summary

A test case is an important software testing document that explains what to test, how to test it, and what result is expected. The key components of a test case include the test case ID, description, preconditions, steps, test data, expected result, actual result, and status. A well-constructed test case is clear, detailed, reproducible, and helps ensure that the software works according to requirements.